

v1.0Deep Dive_ Benevolent Neutralinos

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Neutralino Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0Deep Dive_ Benevolent Neutralinos V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
LHC SUSY exclusion	Pre-Run 3 limits	Run 3 excludes $m_{\tilde{\chi}} > 900$ GeV (wino), > 1100 GeV (higgsino). V2.0 mass range: > 1 TeV only	HIGH	V1.0 simulates now-excluded neutralino masses
Direct detection exclusion	Pre-LZ limits	LZ 2024: all weak-scale neutralinos ($m < 900$ GeV) excluded. V2.0 minimum mass updated	HIGH	V1.0 weak-scale WIMP parameters conflict with LZ results
Majorana fermion data structure	Conceptual description	V2.0: IS_MAJORANA=true prevents creation of separate antiparticle registry entry — particle=antiparticle	MEDIUM	V1.0 could incorrectly pair Majorana neutralino with non-existent antiparticle

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
A-funnel annihilation	General SUSY	V2.0: $m_A \sim 2 \cdot m_{\tilde{\chi}}$ funnel mapped from LHC Higgs data. 10-100 $\times$ enhanced annihilation rate in funnel band	LOW	Enhanced annihilation creates visual 'eruption' event missing from V1.0

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** LHC, Direct
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Deep Dive_ Benevolent Neutralinos V1.0 archived | V2.0 is the active specification as of June 2026